

Deep Double Descent

Where bigger models, longer training, and more data can hurt

COMP34312 | Mathematical Topics in Machine Learning | **Group:** Ayush Kumar | Jacob Yip | Harshita Dassani

Objective	Explain why test error can peak <i>near interpolation</i> and then fall again.
Problem	Classical bias–variance theory predicts one U-shape; modern overparameterised nets interpolate yet generalise.
Approach	Sweep width, training time, samples, noise, optimiser, regularisation; track test <i>and</i> train error.
Contribution	Nakkiran et al. unify model-wise, epoch-wise, and sample-wise double descent via Effective Model Complexity (EMC).

1 From the textbook U-curve to the double-descent peak

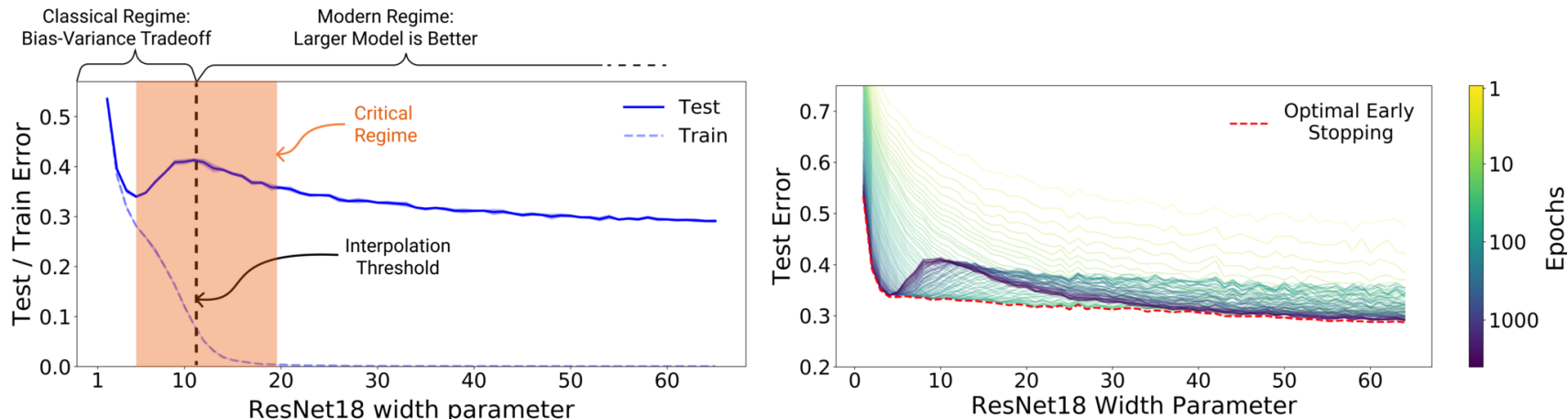


Fig. 1, Nakkiran et al. (2019). Left: test error on ResNet18s, varying width, CIFAR-10 + 15% label noise. Right: test error coloured by training epoch; dashed red curve denotes optimal early stopping.

2 Setting: the lecture decomposition we extend

Following the course notation (Notes §5.1, Eqs. 68–73):

$$\hat{f} \leftarrow \mathcal{A}(f; S), \quad R(f) := \mathbb{E}_{\mathbf{xy}}[\ell(f, \mathbf{y}(\mathbf{x}))], \quad \hat{R}(f, S) := \frac{1}{n} \sum_{i=1}^n \ell(f, \mathbf{y}_i(\mathbf{x}_i))$$

$\hat{f}$ is found by gradient descent (Notes §3.4):

$$\theta_{t+1} = \theta_t - \eta \nabla_{\theta} \hat{R}(\hat{f}_t, S)$$

The excess risk decomposes into three statistically meaningful gaps:

$$\underbrace{R(\hat{f}) - R(y^*)}_{\text{excess risk}} = \underbrace{R(\hat{f}) - R(f_{\text{ERM}})}_{\text{optimisation}} + \underbrace{R(f_{\text{ERM}}) - R(f^*)}_{\text{estimation}} + \underbrace{R(f^*) - R(y^*)}_{\text{approximation}}$$

The classical bias-variance trade-off bundles *approximation + part of estimation* into **bias** and the rest into **variance**; *incomplete*, not wrong. The **paper's pivot**: treat the entire procedure $\mathcal{T}$ – the data-to-classifier procedure parameterised by $\mathcal{F}$, optimiser, training time, regularisation, and label policy – as the single object whose EMC matters.

TERM-BY-TERM BEHAVIOUR AT THE PEAK

Approximation declines with training time as underfitting is dominated by **approximation error**; **estimation** spikes at $\text{EMC} \approx n$; **approximation** declines with hypothesis-class size; All three terms co-exist and interact, i.e. a trade-off.

3 Definition + Hypothesis (Nakkiran et al. 2019)

DEFINITION 1: EFFECTIVE MODEL COMPLEXITY

A **training procedure** $\mathcal{T}$ maps any labelled train set $S = \{(\mathbf{x}_i, \mathbf{y}_i)\}_{i=1}^n$ to a classifier $\mathcal{T}(S)$. For an error tolerance $\varepsilon > 0$,

$$\text{EMC}_{\mathcal{D}, \varepsilon}(\mathcal{T}) := \max \{ n \mid \mathbb{E}_{S \sim \mathcal{D}^n} [\text{Error}_S(\mathcal{T}(S))] \leq \varepsilon \}$$

i.e. the largest sample size on which $\mathcal{T}$ drives train error below ε . Note that EMC depends on the optimiser, training time, augmentation, and labels.

HYPOTHESIS 1: GENERALISED DOUBLE DESCENT

For “natural” $\mathcal{D}$, neural-network procedures $\mathcal{T}$, and small ε (paper: $\varepsilon = 0.1$):

EMC $\ll n$ **Under-parameterised**: any change to $\mathcal{T}$ that *increases* EMC *decreases* test error.

EMC $\approx n$ **Critically parameterised**: the sign of the same change is *indeterminate*.

EMC $\gg n$ **Over-parameterised**: any change to $\mathcal{T}$ that *increases* EMC *decreases* test error.

Why this is a hypothesis, not a theorem. ε and “sufficiently smaller / larger” are heuristic; the critical-interval width is not formally specified for deep nets. **EMC is operational, not predictive**: peak location must be *measured*, not derived.

5 Sample-wise descent: increasing n can locally raise test loss

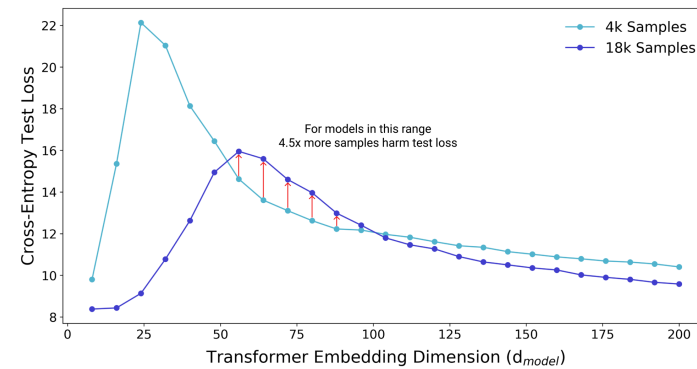


Fig. 3, Nakkiran et al. IWSLT’14 De-En, 6-layer Transformer, $d_{\text{tr}} = 4d_{\text{model}}$, 80k gradient steps, 10% label smoothing. Highlighted bars indicate the d_{model} interval over which a 4.5x increase in n raises test loss.

- Fixing $(\mathcal{T}, f)$ and increasing n leaves the interpolation threshold in EMC space unchanged but shifts n relative to it; widths near the small-sample threshold are thereby displaced into the critical regime.
- The non-monotonicity is local: asymptotically, additional samples reduce empirical risk.
- Practical implication**: an **ensemble of models trained on disjoint sub-samples** sits past each member’s peak; averaging cancels member-specific interpolation noise (Fig. 28).

4 Main empirical evidence: model-wise $\times$ epoch-wise double descent

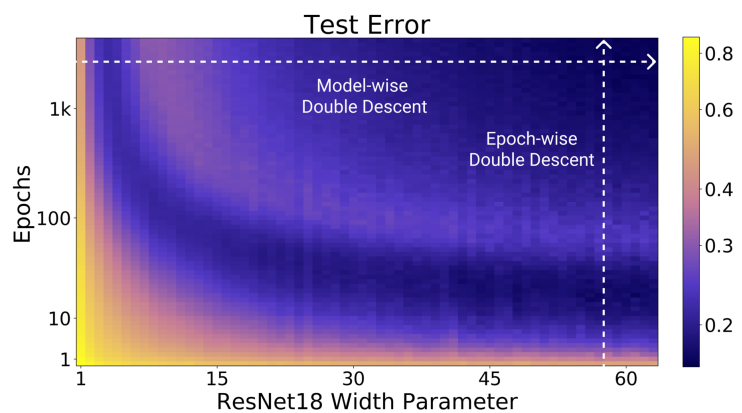


Fig. 2 (left), test error. Test error attains ≈ 0.47 at width $k \approx 10$ and decreases to ≈ 0.24 at $k = 64$.

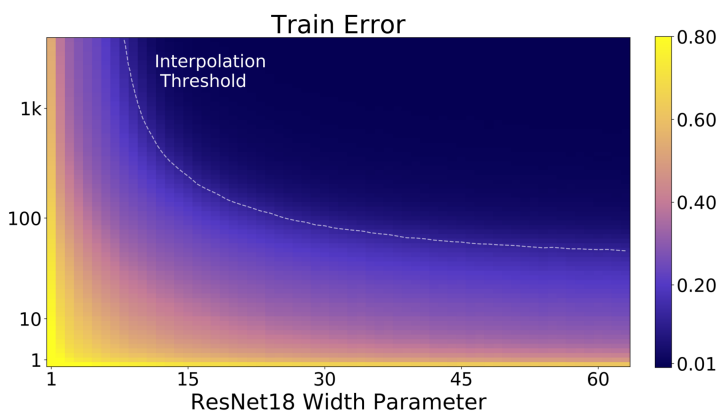


Fig. 2 (right), train error. Dashed curve denotes the interpolation threshold ($\hat{R} \rightarrow 0$).

READING THE HEATMAPS

- Same axes**: ResNet18 width $k \in [1, 64]$ vs training epochs (log).
- Horizontal slice**: **model-wise** peak as width grows. **Vertical slice**: **epoch-wise** peak as training continues.
- Key alignment**: the locus of high test error in the left heatmap coincides with the train-interpolation contour (dashed) in the right heatmap; the peak occurs at the threshold, not past it.

SETUP + KEY NUMBERS (PAPER §4)

ResNet18, CIFAR-10, 15% label noise, augmentation, Adam (lr 1e-4), 4k epochs. Sample-wise (Fig. 3): a **4.5x data increase** (4k→18k) raises test loss for $d_{\text{model}} \in [0, 200]$. Paper uses $\varepsilon = 0.1$.

The unifying variable is EMC: any modification of $\mathcal{T}$ that moves EMC across n shifts the peak correspondingly.

6 Mechanism (rigorous for linear models; conjectural for deep nets)

CRITICALLY PARAMETERISED

The hypothesis class admits effectively a unique interpolant. Constraining it to fit noisy or mis-specified labels perturbs the on-distribution map; test risk consequently spikes.

OVER-PARAMETERISED

A continuum of interpolants exists. The optimiser’s **implicit bias** (e.g. SGD toward minimum-norm) selects an interpolant that absorbs label noise while preserving smoothness on the data manifold.



WHERE THE MECHANISM IS PROVEN

Mei & Montanari (2022, Thm. 2) establish a test-risk peak at $d=n$ for random-features regression under mis-specification; Belkin et al. (2019) prove an analogous peak for least-squares; Bartlett et al. (2020) prove benign overfitting under minimum-norm interpolation. The deep-network analogue remains conjectural.

7 Critical reflection: what we do *not* claim

- The bias-variance trade-off is not invalidated.** It remains correct within a fixed hypothesis class; it is *incomplete* once $\mathcal{T}$ is permitted to vary.
- Model width, training time, and sample size are not monotonic in test risk.** Within the critical interval each can locally *increase* it.
- Label noise is not the cause of the peak.** Noise amplifies it; the peak persists at **0% noise** (CIFAR-100, Fig. 4a; CNN, Fig. 7) under model mis-specification.
- Parameter count does not equal EMC.** Parameters constitute one input to EMC; optimiser, epochs, augmentation, and label policy contribute jointly.

8 Practical implications and open questions

IMPLICATIONS FOR PRACTICE

- Diagnostic.** Train and test curves decouple near the interpolation threshold; monitor both.
- Per-model mitigation.** Weight decay, early stopping, or scaling well beyond the threshold.
- Aggregate mitigation.** Ensemble K sub-sampled models, each over-parameterised relative to its own n (Fig. 28).
- Augmentation.** Data augmentation shifts the peak rightward (Fig. 5) without changing the architecture.

OPEN SCIENTIFIC QUESTIONS

- A principled choice of ε and closed-form critical-interval width.
- A non-linear EMC theorem for over-parameterised networks.
- Characterisation of SGD’s implicit bias for realistic architectures and losses.

Interpolation is a threshold to understand, not a final verdict.

KEY REFERENCES

[1] Nakkiran et al. *Deep Double Descent*. 2019. [2] Belkin et al. *Reconciling ML practice & bias-variance*. 2019. [3] Zhang et al. *Rethinking generalization*. 2017. [4] Mei & Montanari. 2022. [5] Bartlett et al. 2020. [6] Brown & Mukherjee. 2026.

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